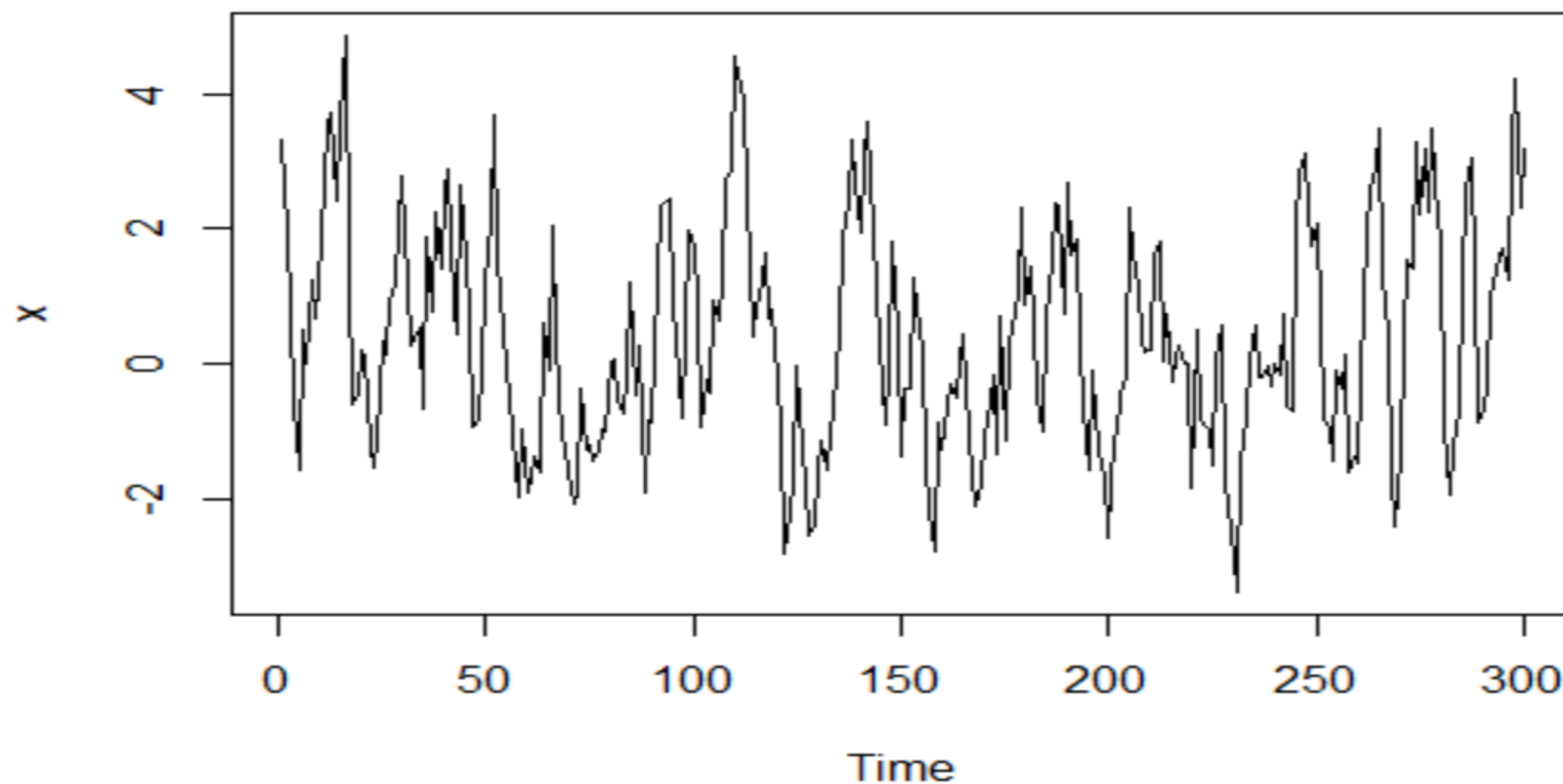


Autocorrelation and Partial Autocorrelation (ACF and PACF)^{[1],[2]}



MA(q)? $q=?$

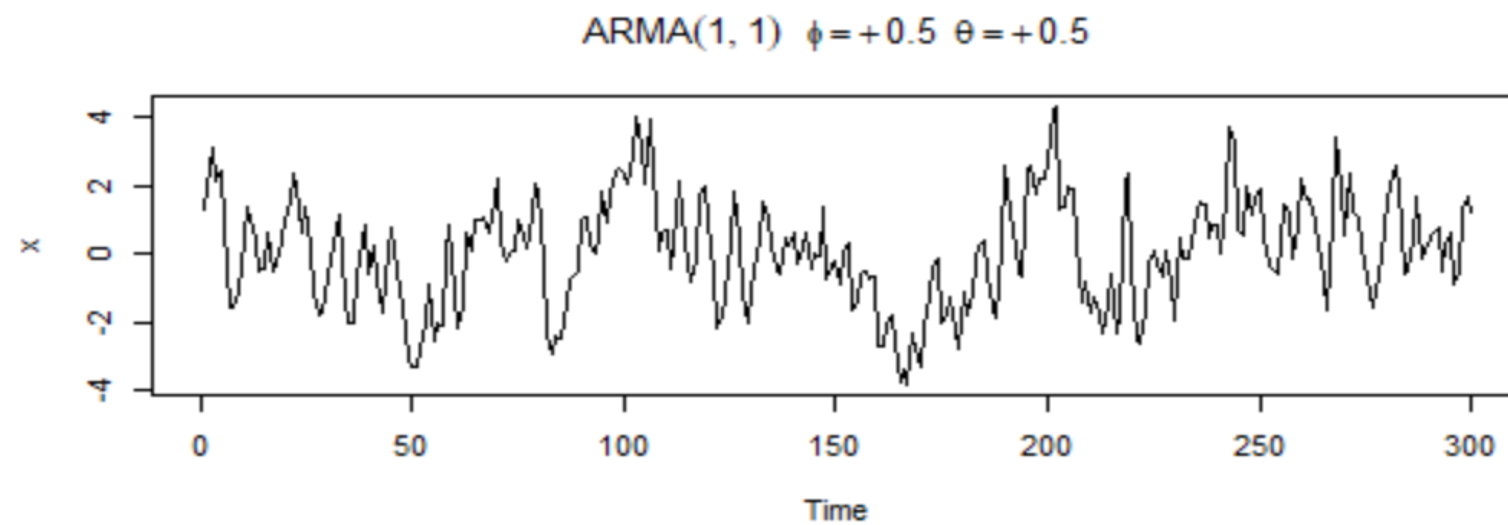
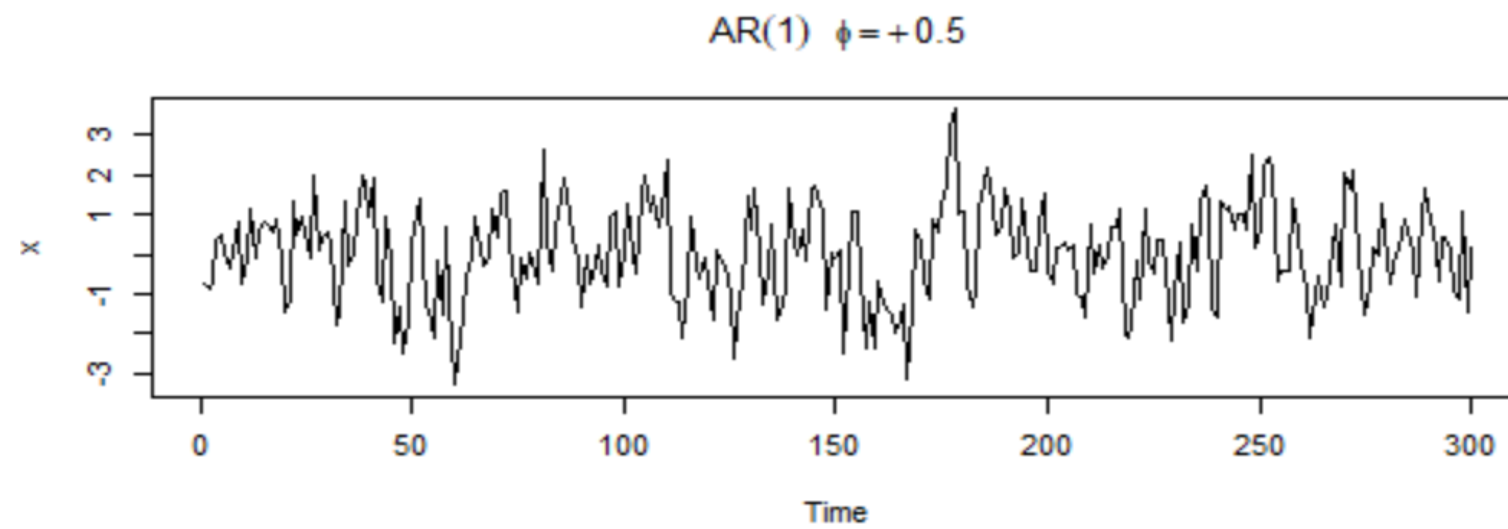
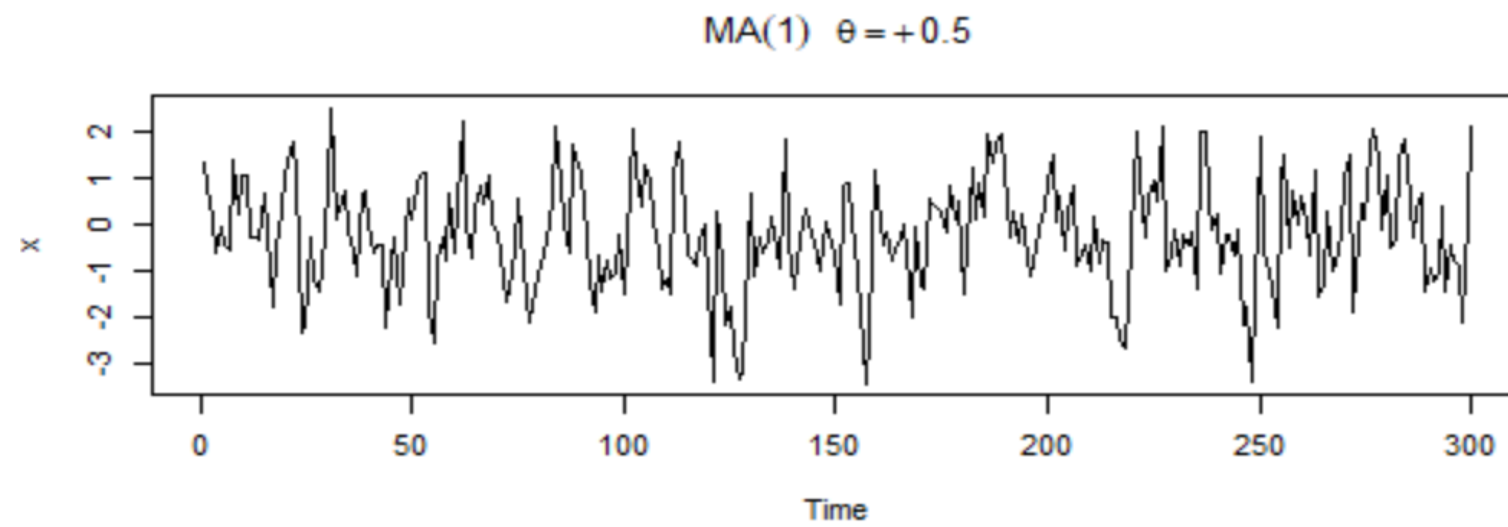
AR(p) $p=?$

ARMA(p,q)?

[1] Robert Shumway, David Stoffer. Time Series Analysis and its Applications with R Examples, 4th edition, Springer, 2017 (chapter 3)

[2] Peter Brockwell, Richard Davis. Time Series: Theory and Methods, Springer-Verlag, 1987 (chapter 3)

ACF and PACF



ACF

1) MA(q) model

$$X_t = \theta(B)W_t, \quad \theta(B) = 1 + \theta_1 B + \cdots + \theta_q B^q, \quad \theta_q \neq 0$$

$$E(X_t) = \sum_{j=0}^q \theta_j E(W_{t-j}) = 0, \quad \theta_0 = 1$$

$$\begin{aligned} \gamma(h) &= \text{cov}(X_{t+h}, X_t) = \text{cov}\left(\sum_{j=0}^q \theta_j W_{t+h-j}, \sum_{k=0}^q \theta_k W_{t-k}\right) \\ &= \begin{cases} \sigma_w^2 \sum_{k=0}^{q-h} \theta_{k+h} \theta_k, & 0 \leq h \leq q \\ 0, & h > q \end{cases} \end{aligned}$$

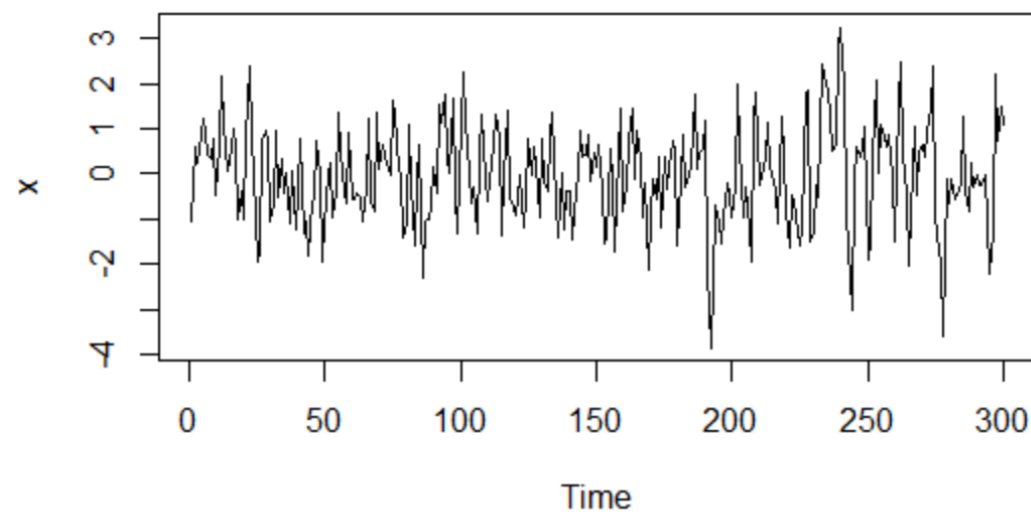
$$\gamma(h) = \gamma(-h)$$

$$\rho(h) = \frac{\gamma(h)}{\gamma(0)}$$

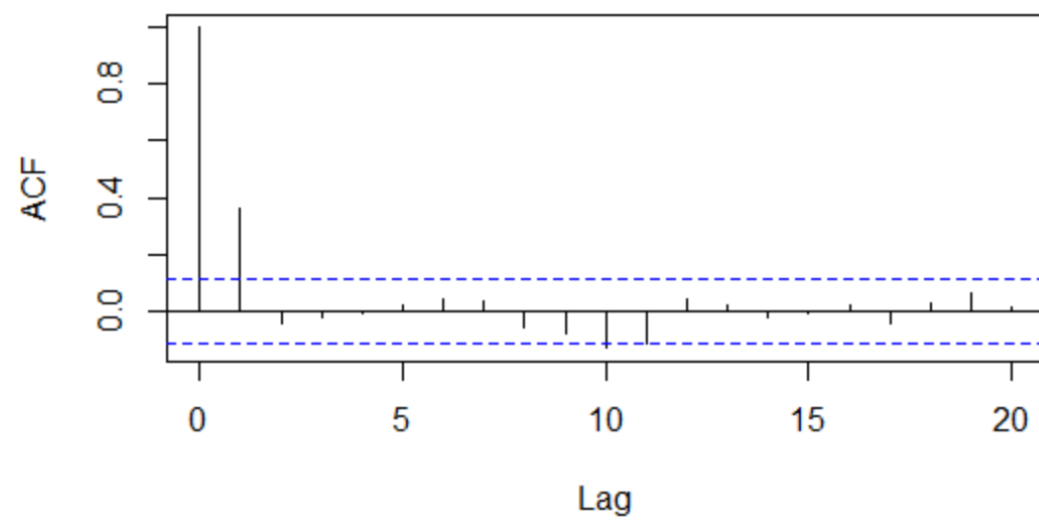
ACF

For an MA(q) process, $\gamma(h)$ is cut off after q lags.

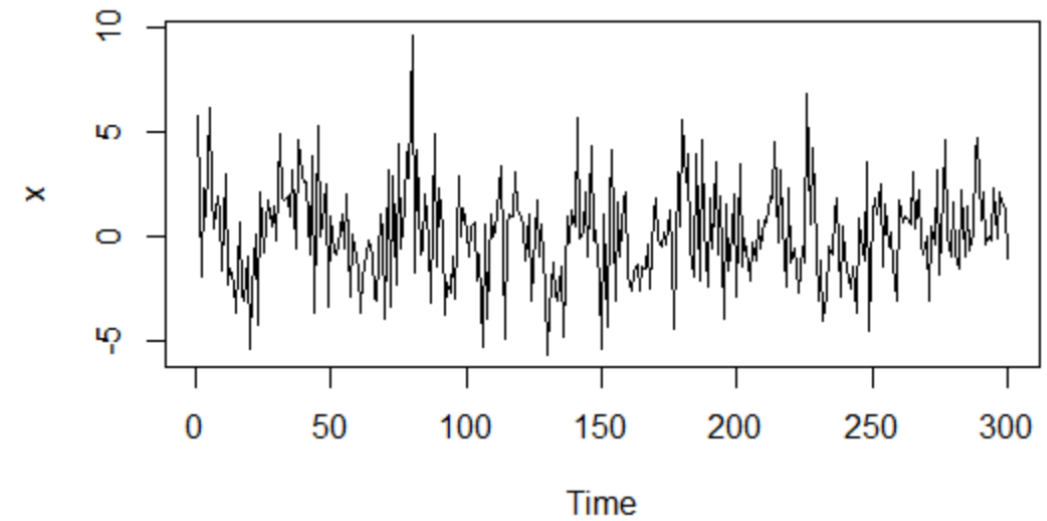
MA(1) $\theta = +0.5$



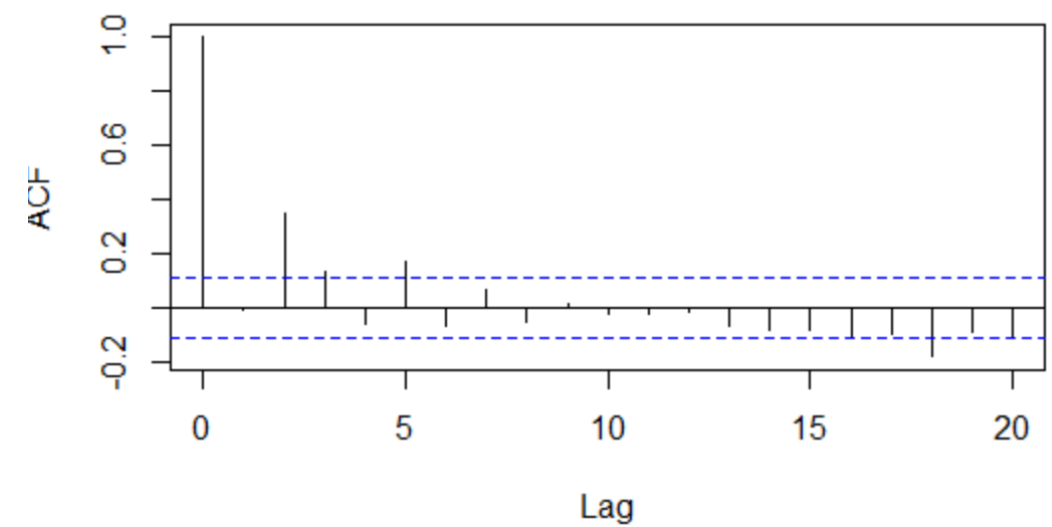
Series x



MA(5) $\theta = (1, 1, 1.4, -1, 1)$



Series x



ACF

2) The causal ARMA (p, q) model

$$\phi(B)X_t = \theta(B)W_t$$

$$E(X_t) - \phi_1 E(X_{t-1}) - \dots - \phi_p E(X_{t-p}) = 0, \quad E(X_t) = \mu$$

$$\left. \begin{array}{l} \mu(1 - \phi_1 - \dots - \phi_p) = 0 \\ \text{the zeros of } \phi(z) \text{ are outside the unit circle} \end{array} \right\} \Rightarrow \mu = 0$$

The process is causal $\Rightarrow X_t = \sum_{j=0}^{\infty} \psi_j W_{t-j}$, $\psi_0 = 1$

$$\phi(B)X_t = \theta(B)W_t \quad | \cdot X_{t-h}$$

ACF

$$\begin{aligned} X_t X_{t-h} - \phi_1 X_{t-1} X_{t-h} - \cdots - \phi_p X_{t-p} X_{t-h} \\ = W_t X_{t-h} + \theta_1 W_{t-1} X_{t-h} + \cdots + \theta_q W_{t-q} X_{t-h} \quad | \text{ apply } E \end{aligned}$$

$$\gamma(h) - \phi_1 \gamma(h-1) - \cdots - \phi_p \gamma(h-p) = E(W_t X_{t-h}) + \cdots + E(\theta_q W_{t-q} X_{t-h})$$

$$X_{t-h} = \sum_{j=0}^{\infty} \psi_j W_{t-h-j} = \sum_{k=h}^{\infty} \psi_{k-h} W_{t-k}$$

$$E(\theta_j W_{t-j} X_{t-h}) = \theta_j E(W_{t-j} \sum_{k=h}^{\infty} \psi_{k-h} W_{t-k}) = \theta_j \psi_{j-h} \sigma_w^2, \quad h \leq j \leq q$$

ACF

We get the difference equations:

$$\gamma(h) - \phi_1\gamma(h-1) - \dots - \phi_p\gamma(h-p) = 0, \quad h \geq \max(p, q+1)$$

with initial conditions:

$$\gamma(h) - \phi_1\gamma(h-1) - \dots - \phi_p\gamma(h-p) = \sigma_w^2 \sum_{j=h}^q \theta_j \psi_{j-h}, \quad 0 \leq h < \max(p, q+1).$$

[for the closed form solution, see Peter Brockwell, Richard Davis – Time series: Theory and Methods, Springer-Verlag, 1987- pag. 91-]

ACF

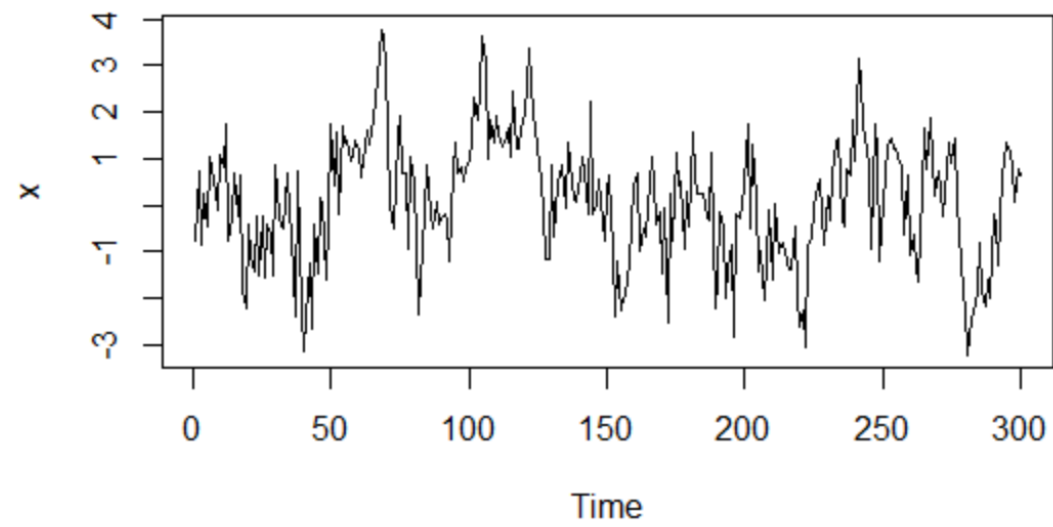
3) The causal AR(p) model

$$\gamma(h) = \sum_{i=1}^p \phi_i \gamma(h-i), \quad h > 0$$
$$\gamma(0) = \sum_{i=1}^p \phi_i \gamma(i) + \sigma_w^2$$

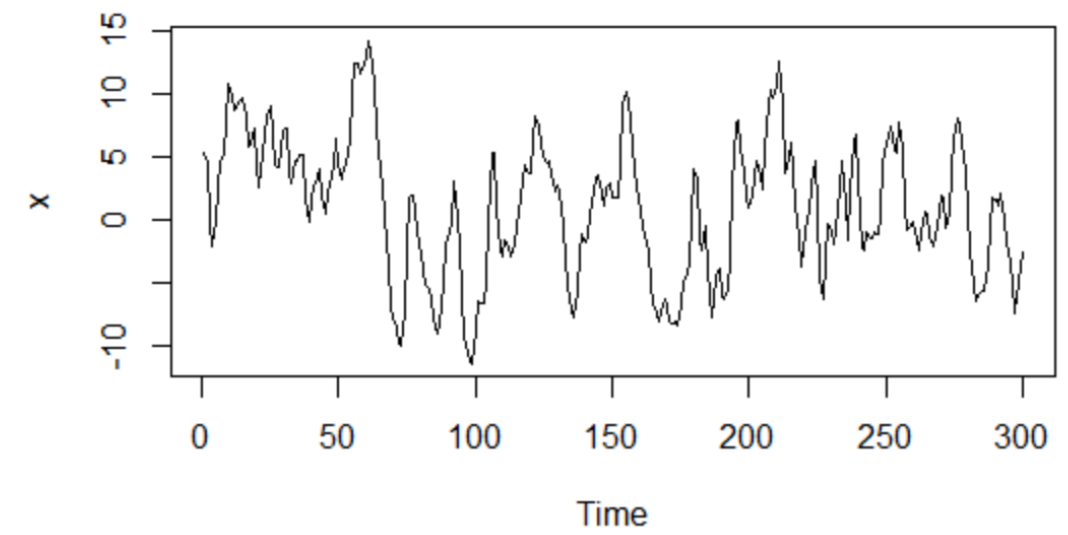
With the ACF it is unlikely to see the difference between an ARMA model and an AR model.

ACF

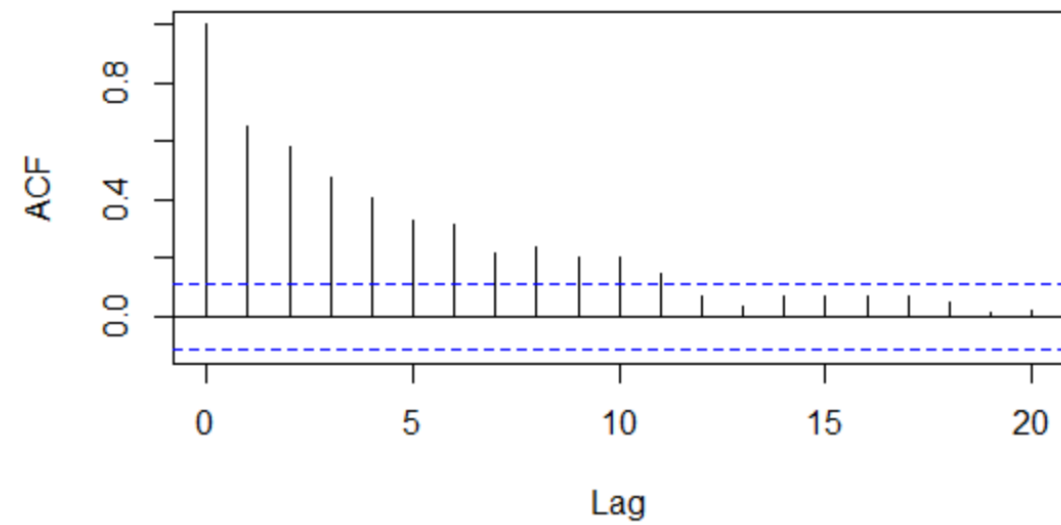
AR(2) $\phi = (.5, .2)$



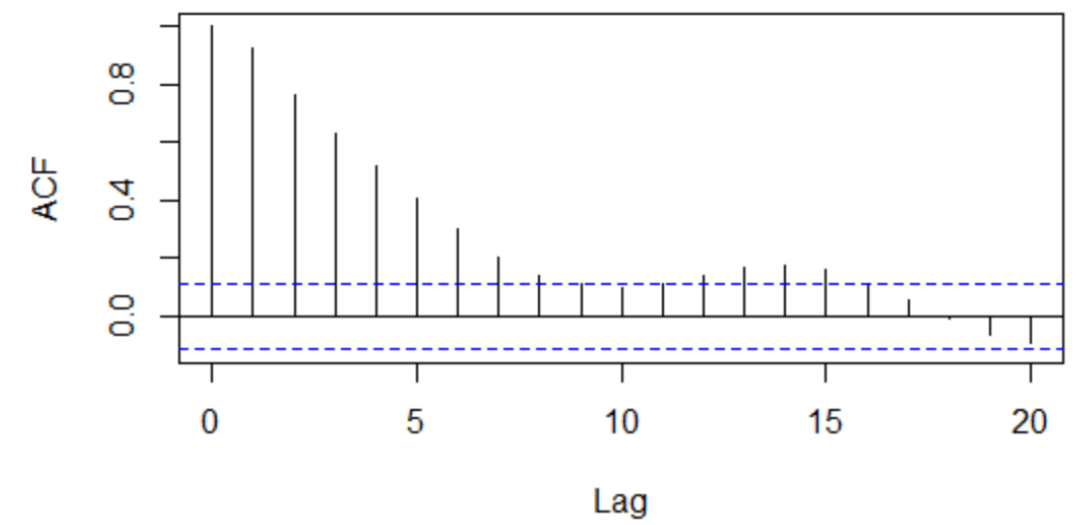
ARMA(2, 2) $\phi = (.5, .2)$ $\theta = (2, 1)$



Series x



Series x



PACF

It removes the linear effects of $X_1, \dots, X_{h-1}$ in the correlation between X_h and X_0 .

$$\dots X_0, \underbrace{X_1, \dots, X_{h-1}}_{\text{“partialled out”}}, X_h, \dots$$

Consider the AR(1) model:

$$X_t = \phi X_{t-1} + W_t$$

$$\begin{aligned}\gamma(2) &= \text{cov}(X_t, X_{t-2}) = \text{cov}(\phi X_{t-1} + W_t, X_{t-2}) \\ &= \text{cov}(\phi^2 X_{t-2} + \phi W_{t-1} + W_t, X_{t-2}) = \phi^2 \gamma(0)\end{aligned}$$

We can partial out the effect of X_{t-1} in the dependence of X_t on X_{t-2} :

$$\text{cov}(X_t - \phi X_{t-1}, X_{t-2} - \phi X_{t-1}) = \text{cov}(W_t, X_{t-2} - \phi X_{t-1}) = 0$$

PACF

A *Hilbert* space $\mathcal{H}$ is an inner-product space which is complete, i.e., every *Cauchy* sequence $\{X_n\}$ (that is $\|X_n - X_m\| \rightarrow 0$ as $m, n \rightarrow \infty$) converges in norm to $X \in \mathcal{H}$:

$$\|X_n - X\| \rightarrow 0 \text{ as } n \rightarrow \infty$$

Example - the space $L^2(\Omega, \mathcal{K}, \mathbb{P})$ with the inner product

$$\langle X, Y \rangle = E(XY) \text{ and } E(X^2) < \infty.$$

The closed span of $\{X_1, \dots, X_k\}$ is the set of all linear combinations

$$\alpha_1 X_1 + \dots + \alpha_k X_k, \quad \alpha_1, \dots, \alpha_k \in \mathbb{R}.$$

PACF

The best linear predictor of X in $\overline{sp}\{X_1, \dots, X_k\}$ is

$$P_{\overline{sp}\{X_1, \dots, X_k\}} X = \sum_{i=1}^k \alpha_i X_i,$$

where $\alpha_1, \dots, \alpha_k$ satisfy:

$$\sum_{i=1}^n \alpha_i E(X_i X_j) = E(X X_j), \quad j = 1, \dots, n$$

Also, $\alpha_1, \dots, \alpha_k$ minimize $E[(X - \sum_{i=1}^k \alpha_i X_i)^2]$ (by the projection theorem)

[for more, see Chapter 2 in Brockwell & Davis]

PACF

Def. The partial autocorrelation function (PACF) of a stationary process X_t , denoted ϕ_{hh} , is defined as:

$$\begin{aligned}\phi_{11} &= \text{corr}(X_{t+1}, X_t) = \rho(1) \\ \phi_{hh} &= \text{corr}(X_{t+h} - \hat{X}_{t+h}, X_t - \hat{X}_t), \quad h \geq 2,\end{aligned}$$

where

$$\begin{aligned}\hat{X}_{t+h} &= P_{\overline{sp}\{X_{t+1}, \dots, X_{t+h-1}\}} X_{t+h} \\ \hat{X}_t &= P_{\overline{sp}\{X_{t+1}, \dots, X_{t+h-1}\}} X_t\end{aligned}$$

PACF

Example: The PACF of an AR(1)

$$X_t = \phi X_{t-1} + W_t, \quad |\phi| < 1$$

$$\phi_{11} = \rho(1) = \phi$$

$$\phi_{22} = \text{corr}(X_{t+2} - \hat{X}_{t+2}, X_t - \hat{X}_t)$$

$$\hat{X}_{t+2} = \alpha X_{t+1}$$

$$\begin{aligned} \alpha \text{ minimizes } E[(X_{t+2} - \hat{X}_{t+2})]^2 &= E[(X_{t+2} - \alpha X_{t+1})^2] \\ &= \gamma(0) - 2\alpha\gamma(1) + \alpha^2\gamma(0) \end{aligned}$$

$$\frac{\partial E}{\partial \alpha} = 0 \Rightarrow \alpha = \frac{\gamma(1)}{\gamma(0)} = \rho(1) = \phi$$

$$\hat{X}_t = \beta X_{t+1} \text{ so that } \beta \text{ minimizes } E[(X_t - \hat{X}_t)^2] \Rightarrow \beta = \phi$$

$$\phi_{22} = \text{corr}(X_{t+2} - \phi X_{t+1}, X_t - \phi X_{t+1}) = \text{corr}(W_{t+2}, X_t - \phi X_{t+1}) = 0$$

PACF

The PACF of an AR(p)

$$X_{t+h} = \sum_{j=1}^p \phi_j X_{t+h-j} + W_{t+h}$$

$$\phi_{hh} = \text{corr}(X_{t+h} - \hat{X}_{t+h}, X_t - \hat{X}_t)$$

$$\hat{X}_t = P_{\overline{sp}\{X_{t+1}, \dots, X_{t+h-1}\}} X_t$$

For $h > p$, it can be proven that

$$\hat{X}_{t+h} = \sum_{j=1}^p \phi_j X_{t+h-j} \Rightarrow \phi_{hh} = \text{corr}(W_{t+h}, X_t - \hat{X}_t) = 0$$

because $X_t - \hat{X}_t$ depends only on $W_{t+h-1}, W_{t+h-2} \dots$

PACF

The PACF of an invertible MA(q)

$$X_t = -\sum_{j=1}^{\infty} \Pi_j X_{t-j} + W_t$$

Because of the infinite AR representation, the PACF will never cut off, as for AR(p) case.

PACF

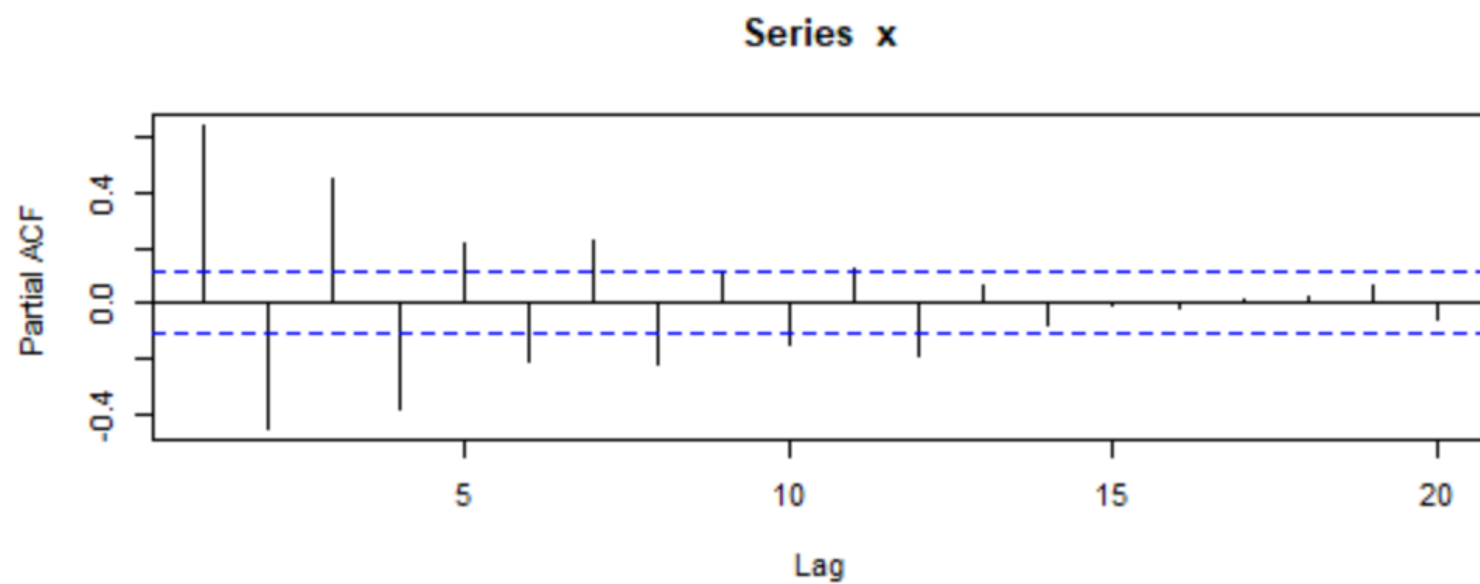
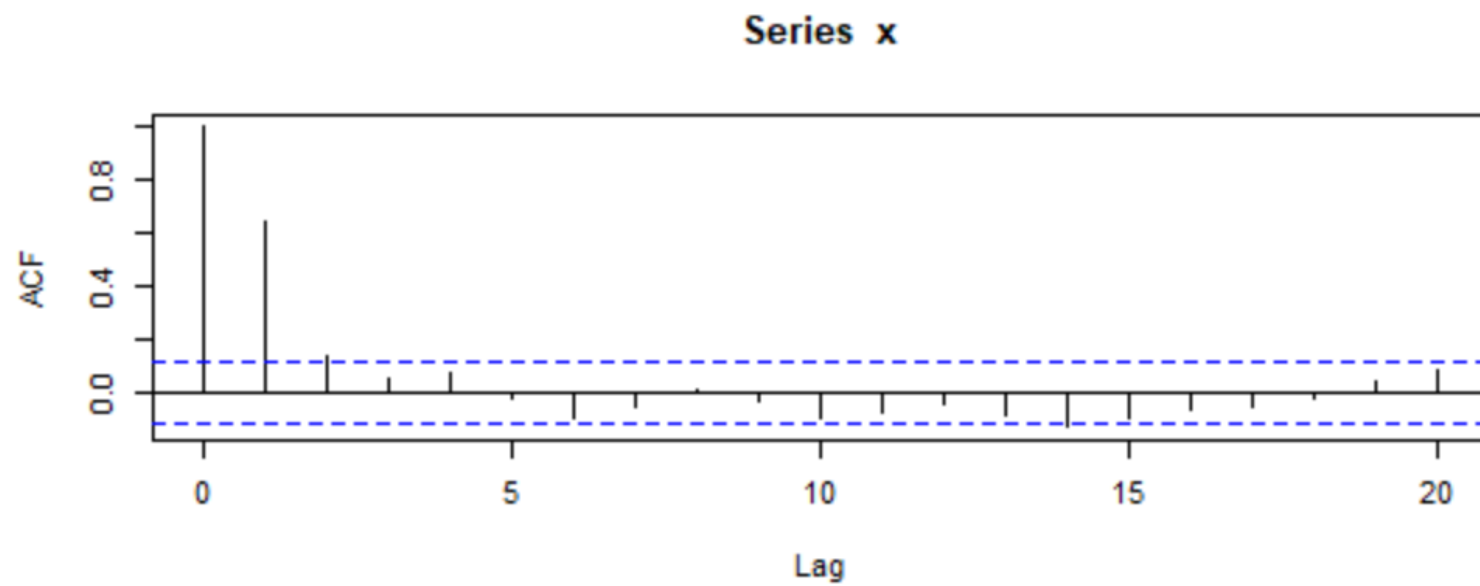
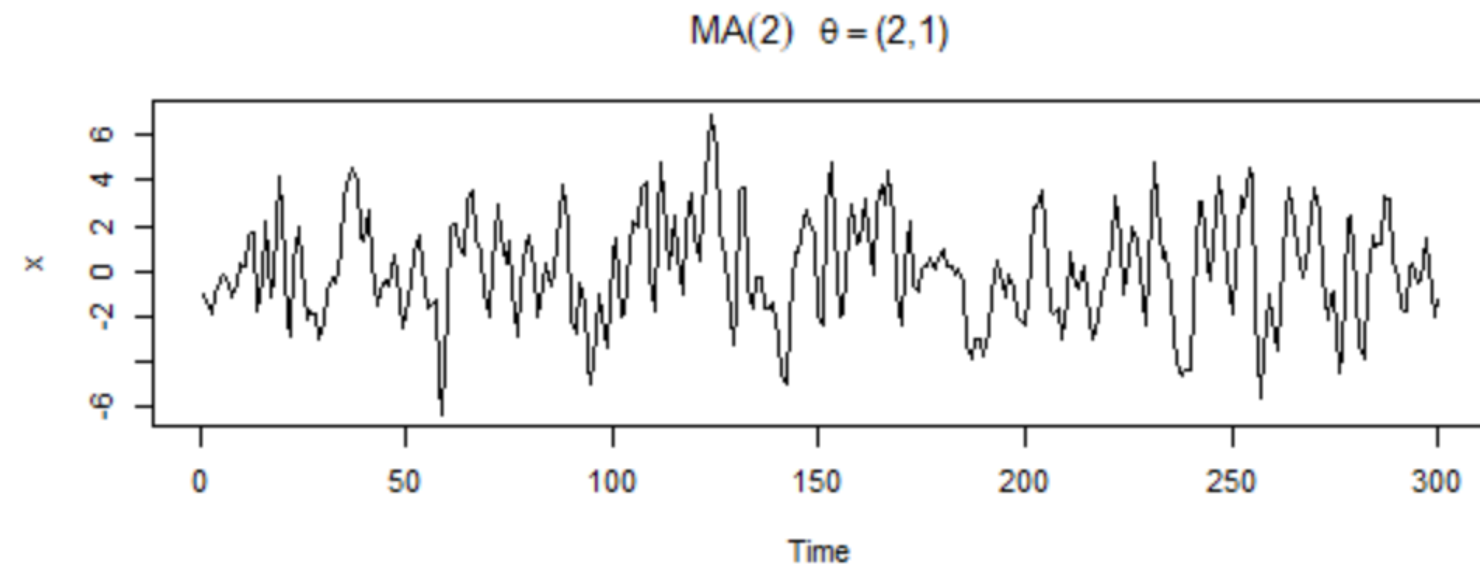
The PACF of an invertible MA(q)

$$X_t = -\sum_{j=1}^{\infty} \Pi_j X_{t-j} + W_t$$

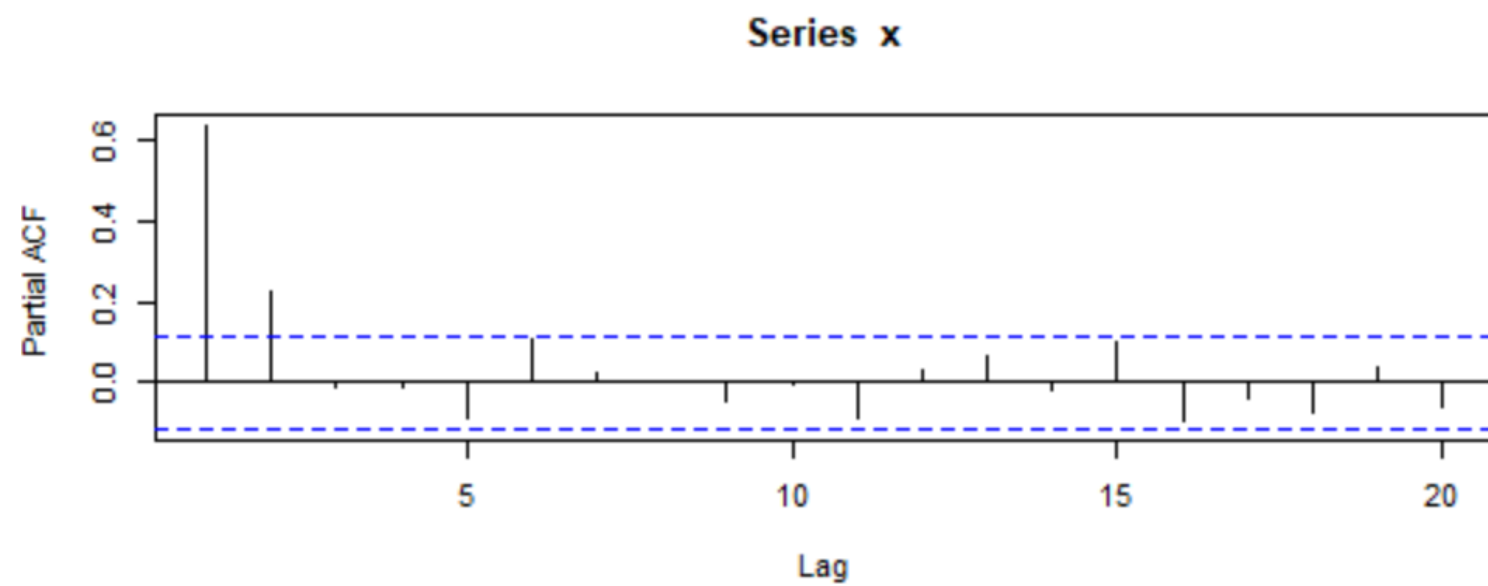
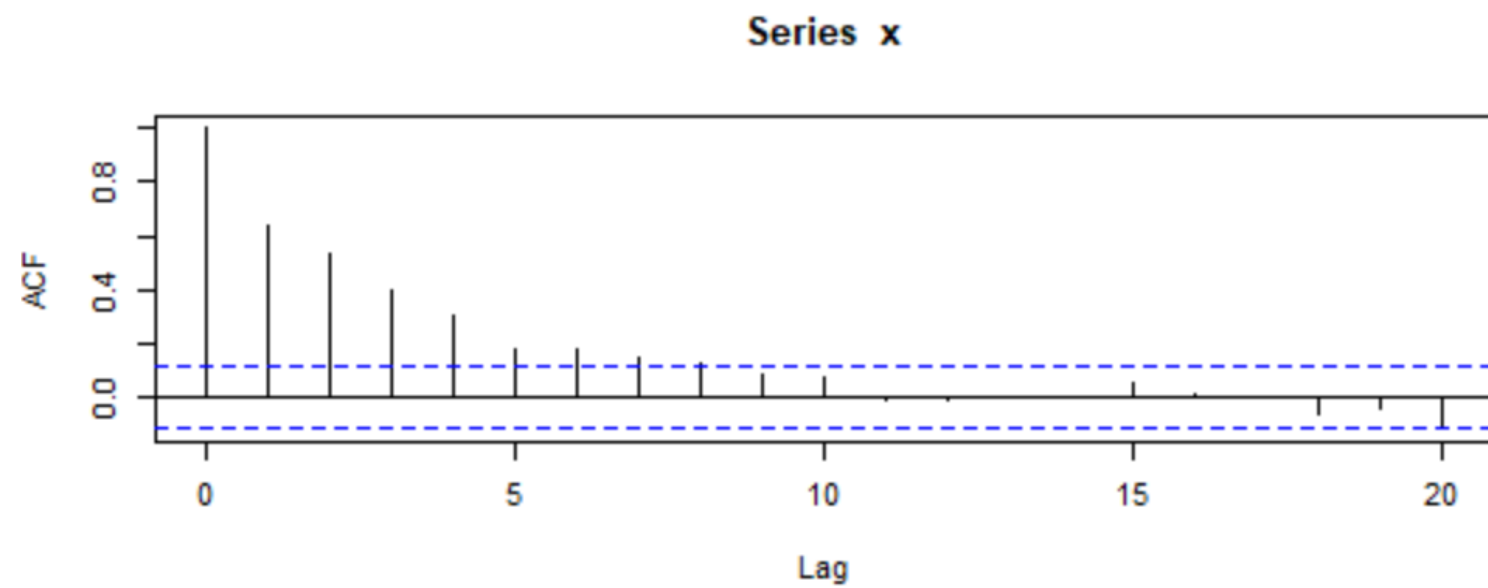
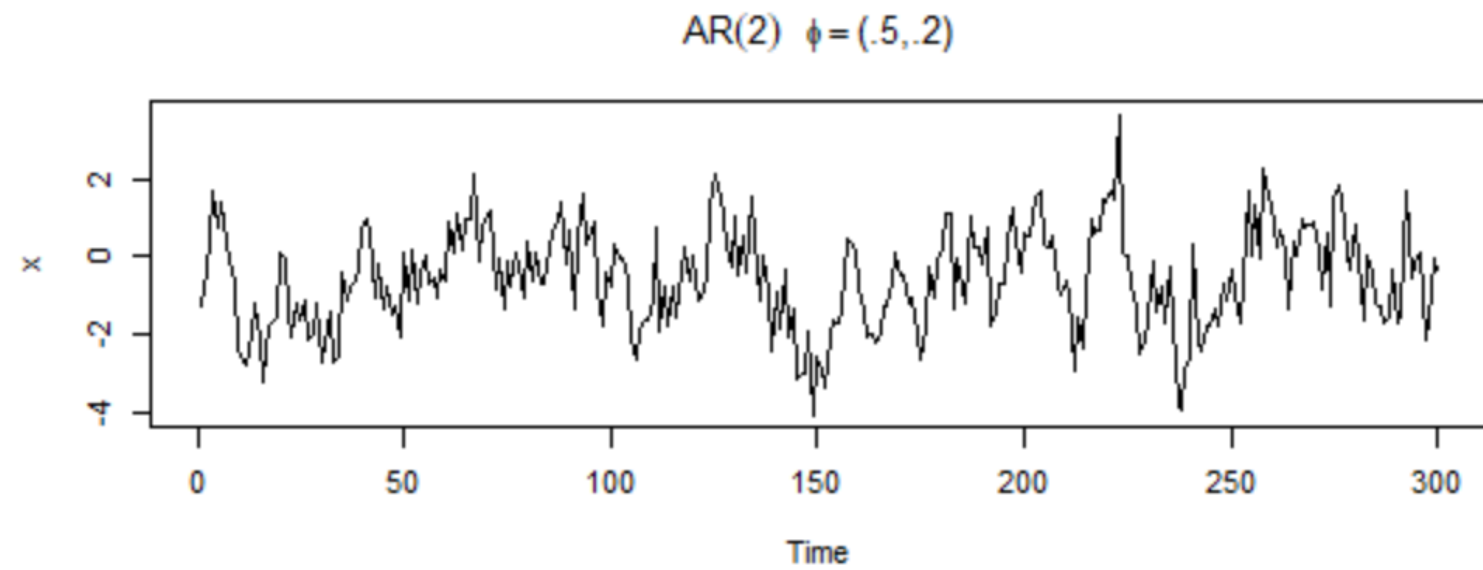
Because of the infinite AR representation, the PACF will never cut off, as for AR(p) case.

	AR(p)	MA(q)	ARMA (p, q)
ACF	Decays	Cuts off after lag q	Decays
PACF	Cuts off after lag p	Decays	Decays

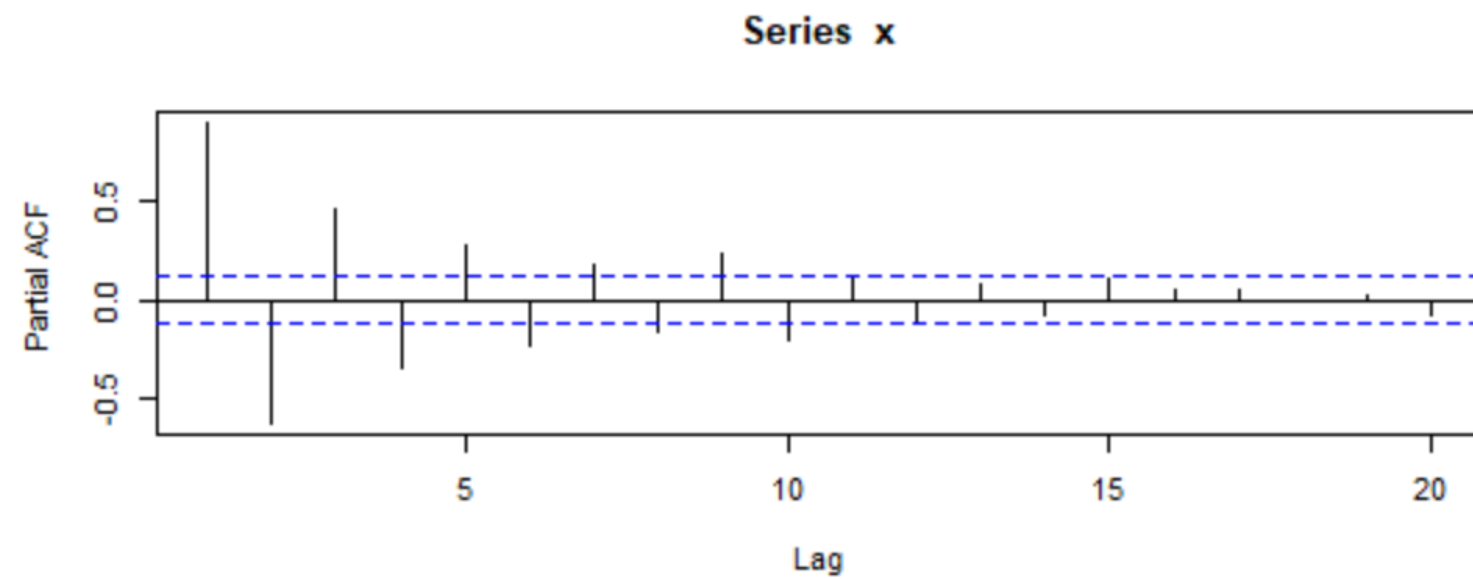
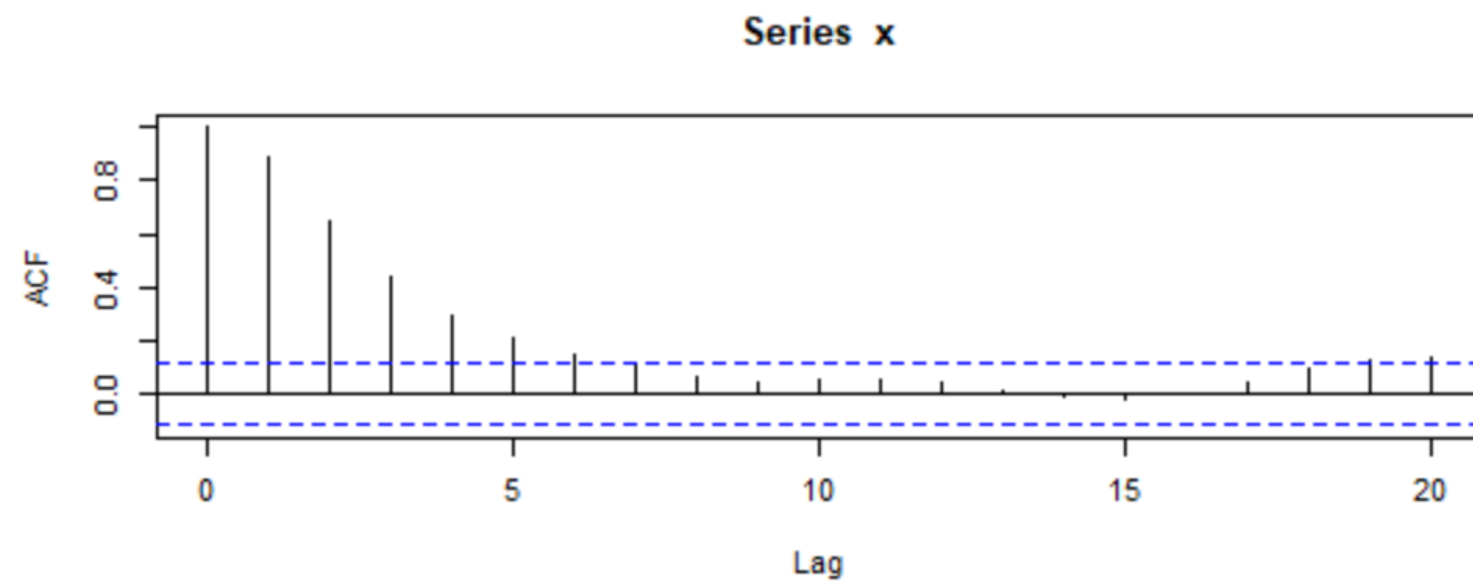
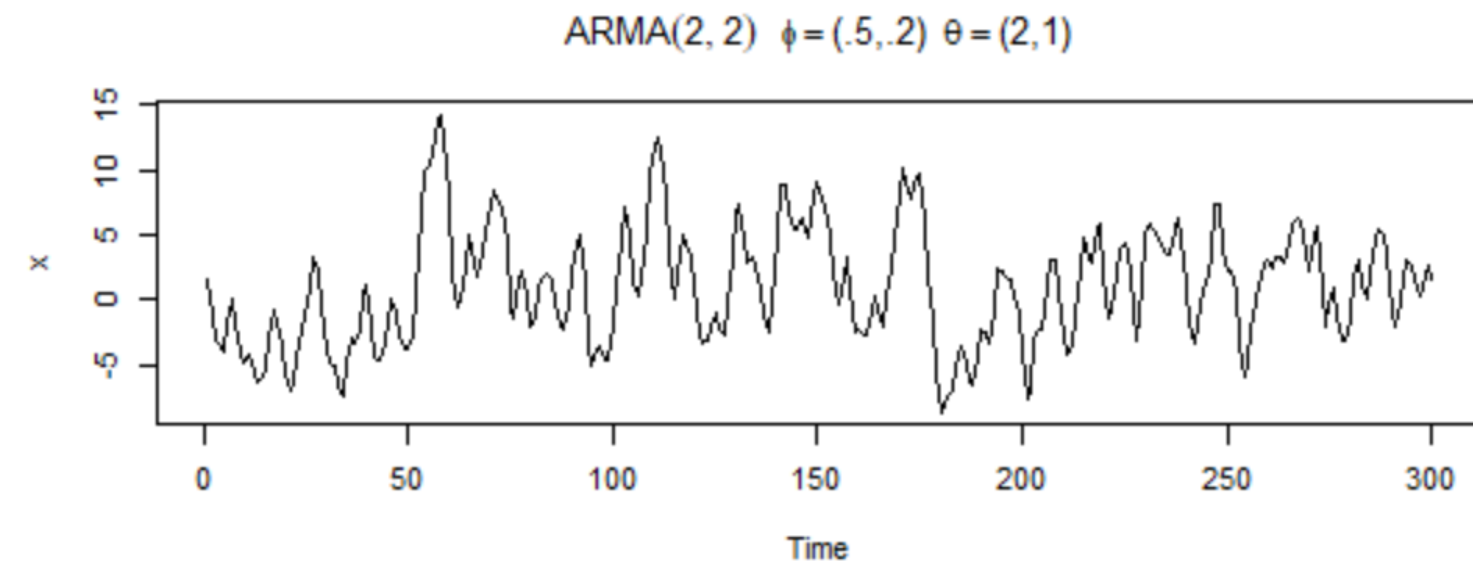
PACF



PACF



PACF



Forecasting^{[1],[2]}

It's difficult to make predictions, especially about the future.

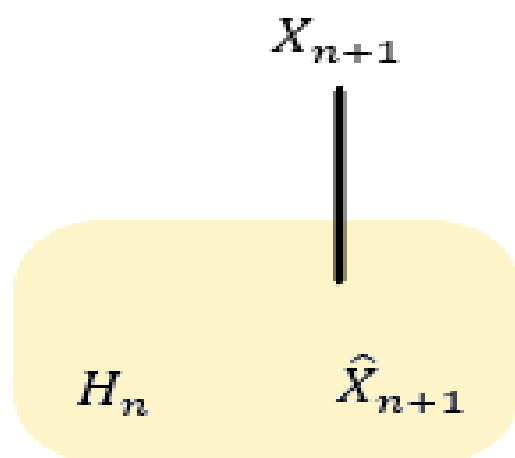
Danish proverb? Yogi Berra?

Equations for one-step predictors

We assume that X_t is stationary and the covariance function is known.

Let $H_n = \overline{sp}\{X_1, \dots, X_n\}$

$\hat{X}_{n+1} = P_{H_n} X_{n+1}$ – *the Best Linear Predictor (BLP) of X_{n+1}*



$$\hat{X}_{n+1} = \phi_{n1}X_n + \dots + \phi_{nn}X_1, \quad n \geq 1,$$

$$\text{where } E[(X_{n+1} - \sum_{j=1}^n \phi_{nj}X_{n+1-j})X_{n+1-k}] = 0, \quad k = 1, \dots, n$$

[1] Robert Shumway, David Stoffer. Time Series Analysis and its Applications with R Examples, 4th edition, Springer, 2017 (chapter 3)

[2] Peter Brockwell, Richard Davis. Time Series: Theory and Methods, Springer-Verlag, 1987 (chapter 5)

Forecasting

It can be rewritten as

$$\sum_{j=1}^n \phi_{nj} \gamma(k-j) = \gamma(k), \quad k = 1, \dots, n$$

Or, in a matrix form:

$$\Gamma_n \phi_n = \gamma_n,$$

where

$$\Gamma_n = \begin{pmatrix} \gamma_{(0)} & \gamma_{(1)} & \cdots & \gamma_{(n-1)} \\ \vdots & & \ddots & \vdots \\ \gamma_{(n-1)} & & \cdots & \gamma_{(0)} \end{pmatrix} \quad \phi_n = \begin{pmatrix} \phi_{n1} \\ \vdots \\ \phi_{nn} \end{pmatrix} \quad \gamma_n = \begin{pmatrix} \gamma_{(1)} \\ \vdots \\ \gamma_{(n)} \end{pmatrix}$$

Forecasting

Proposition: If $\sigma_w^2 > 0$ and $\gamma(h) \rightarrow 0$ as $h \rightarrow \infty$, then Γ_n is non-singular for every n .

$\phi_n = \Gamma_n^{-1} \gamma_n$ - the elements of ϕ_n are unique.

The predictor $\hat{X}_{n+1} = \sum_{j=1}^n \phi_{nj} X_{n+1-j}$, $n = 1, 2, \dots$

Forecasting

The mean squared error is:

$$V_n = E \left[(X_{n+1} - \hat{X}_{n+1})^2 \right] = E[(X_{n+1} - \phi_n' X)^2],$$

where $X = (X_n, X_{n-1}, \dots, X_1)'$.

$$\begin{aligned} V_n &= E[(X_{n+1} - \gamma_n' \Gamma_n^{-1} X)^2] \\ &= E(X_{n+1}^2 - 2 \gamma_n' \Gamma_n^{-1} X X_{n+1} + \gamma_n' \Gamma_n^{-1} X X' \Gamma_n^{-1} \gamma_n) \\ &= \gamma(0) - 2 \gamma_n' \Gamma_n^{-1} \gamma_n + \gamma_n' \Gamma_n^{-1} \Gamma_n \Gamma_n^{-1} \gamma_n \\ &= \gamma(0) - \gamma_n' \Gamma_n^{-1} \gamma_n. \end{aligned}$$

ϕ_n and V_n can be computed iteratively, due to Levinson&Durbin.

Forecasting

The Durbin-Levinson algorithm

$$\text{initialization: } \begin{cases} \phi_{11} = \frac{\gamma(1)}{\gamma(0)} \\ V_0 = \gamma(0) \end{cases}$$
$$\phi_{nn} = \frac{\gamma(n) - \sum_{k=1}^{n-1} \phi_{n-1,k} \gamma(n-k)}{V_{n-1}}, n \geq 1$$

$$\phi_{nk} = \phi_{n-1,k} - \phi_{nn} \phi_{n-1,n-k}, \quad k = 1, \dots, n-1, \quad n \geq 2$$

$$V_n = V_{n-1} [1 - \phi_{nn}^2]$$

Order of computation:

$$\phi_{11}, V_1, \phi_{22}, \phi_{21}, V_2, \phi_{33}, \phi_{31}, \phi_{32}, V_3 \dots$$

The predictor: $\hat{X}_{n+1} = \phi_{n1}X_n + \phi_{n2}X_{n-1} + \dots + \phi_{nn}X_1$

Forecasting

Proposition: The PACF of a stationary process $\{X_t\}$ can be obtained from the Durbin-Levinson algorithm, as ϕ_{nn} for $n = 1, 2, \dots$

Another useful algorithm for calculating the forecast $\hat{X}_{n+1}$ is based on the innovations $X_i - \hat{X}_i$, $i = \overline{1, n}$.

Forecasting

The innovations algorithm

The predictor: $\hat{X}_{n+1} = \sum_{j=1}^n \theta_{nj} (X_{n+1-j} - \hat{X}_{n+1-j})$, $n \geq 1$

V_n – mean squared errors

$$\text{initialization: } \begin{cases} V_0 = \gamma(0) \\ \hat{X}_1 = 0 \end{cases}$$

$$\theta_{n,n-k} = \frac{\gamma(n-k) - \sum_{j=0}^{k-1} \theta_{k,k-j} \theta_{n,n-j} V_j}{V_k}, \quad k = 0, 1, \dots, n-1$$

$$V_n = \gamma(0) - \sum_{j=0}^{n-1} \theta_{n,n-j}^2 V_j$$

Order of computation:

$$V_0, \hat{X}_1; \theta_{11}, V_1, \hat{X}_2; \theta_{22}, \theta_{21}, V_2, \hat{X}_3; \theta_{33}, \theta_{32}, \theta_{31}, V_3, \hat{X}_4 \dots$$

Forecasting

Obs. The Durbin-Levinson algorithm and the Innovations algorithm can be adapted to compute the h -step predictors.

$$P_{H_n} X_{n+h} = \phi_{n1}^{(h)} X_n + \cdots + \phi_{nn}^{(h)} X_1, \quad n, h \geq 1$$

[For details see Brockwell & Davis, pages 161, 167].

Forecasting

- 1) Apply the Durbin-Levinson algorithm for an AR(2) process for the first three iterations $n = 1, 2, 3$.
Indicate the first three values of the PACF.

Hint: for an AR (2), recall that:

$$\rho(h) - \phi_1\rho(h-1) - \phi_2\rho(h-2) = 0, \quad \forall h \geq 1$$

and use it for $h=1, 2, 3$ in the D-L algorithm.

Forecasting

2) Apply the Innovations algorithm for an MA(1) process:

$$X_t = \theta W_{t-1} + W_t$$

$$\gamma(0) = (1 + \theta^2)\sigma_w^2$$

$$\gamma(1) = \theta\sigma_w^2$$

$$\gamma(h) = 0, \text{ for } h > 1$$

$\Rightarrow \gamma(n - k) \neq 0$ only for $k = n - 1$ in the formula for $\theta_{n,n-k}$

Forecasting

Thus,

$$\theta_{1,1} = \gamma(1)/V_0$$

$$\theta_{2,2} = 0, \theta_{2,1} = \gamma(1)/V_1$$

$$\theta_{3,3} = 0, \theta_{3,2} = 0, \theta_{3,1} = \gamma(1)/V_2$$

...

$$\theta_{nn} = \theta_{n,n-1} = \dots = \theta_{n,2} = 0, \theta_{n,1} = \gamma(1)/V_n$$

$$V_0 = (1 + \theta^2) \sigma_w^2$$

...

$$V_n = [1 + \theta^2 - \theta^2 \sigma^2 / V_{n-1}] \sigma_w^2$$

$$\hat{X}_{n+1} = \theta(X_n - \hat{X}_n) \sigma_w^2 / V_{n-1}$$

Forecasting

The D-L algorithm is convenient for $AR(p)$ models.

The Innovations algorithm is convenient for $MA(q)$ models.